Arjuna JEE 2027

Maths by Amarnath Anand Sir

Quadratic Equations

DPP: 8

Total Duration - 45 Min

Q1 If $P(x)=x^2-(2-p)x+p-2$ assumes both positive and negative value, then the complete set of values of $'p'$ is-

(A) $(-\infty, 2)$

(B) $(6, \infty)$

(C) $(2,6)$

(D) $(-\infty, 2) \cup(6, \infty)$

Q2 Let $g(x)=x^2-(b+1)x+(b-1)$, where b is a real parameter. The largest natural number b satisfying $g(x)>-2 \forall x \in \mathrm{R}$, is

(A) 1 (B) 2

(C) 3 (D) 4

Q3 For $x \in[1,5], y=x^2-5x+3$ has

(A) least value of $=-1.5$

(B) greatest value of $=3$

(C) least value $=-3.25$

(D) greater value $=\frac{5+\sqrt{13}}{2}$

Q4 If $y=-2x^2-6x+9$, then

(A) Maximum value of y is -11 and it occurs at $x=2$

(B) Minimum value of y is -11 and it occurs at $x=2$

(C) Maximum value of y is 13.5 and it occurs at $x=-1.5$

(D) Minimum value of y is 13.5 and it occurs at $x=-1.5$

Q5 Let r_1, r_2, r_3, be roots of equation $x^3-2x^2+4x+5074=0$, then the value of $\left(r_1+2\right)\left(r_2+2\right)\left(r_3+2\right)$ is

(A) 5050 (B) -5050

(C) -5066 (D) -5068

Q6 Let a, b, c are roots of equation $x^3+8x+1=0$, then the value of $\frac{bc}{(8b+1)(8c+1)}+\frac{ac}{(8a+1)(8c+1)}+\frac{ab}{(8a+1)(8b+1)}$ is equal to

(A) 0

(B) -8

(C) -16

(D) 16

Q7 If $\alpha, \beta, \gamma, \delta$ are the roots of equation $x^4-\beta x+3=0$, then the equation whose solutions are $\frac{\alpha+\beta+\gamma}{\delta^2}, \frac{\alpha+\beta+\delta}{\gamma^2}, \frac{\alpha+\gamma+\delta}{\beta^2}, \frac{\beta+\gamma+\delta}{\alpha^2}$ is-

(A) $3x^4-\beta x^3-1=0$

(B) $3x^4-\beta x^3+1=0$

(C) $3x^4+\beta x^3-1=0$

(D) $3x^4+\beta x^3+1=0$

Q8 If α, β, γ are the roots of the equation $x^3+4x+1=0$, then $(\alpha+\beta)^{-1}+(\beta+\gamma)^{-1}+(\gamma+\alpha)^{-1}$ is

(A) 2 (B) 3

(C) 4 (D) 5

Q9 If the sum of two roots of the equation $x^3-px^2+qx-r=0$ is zero, then

(A) $pq=r$ (B) $qr=p$

(C) $pr=q$ (D) $pqr=1$

Q10 If α, β, γ are the roots of the equation $x^3+ax+b=0$, then $\frac{\alpha^3+\beta^3+\gamma^3}{\alpha^2+\beta^2+\gamma^2}=$

(A) $\frac{3b}{2a}$

(B) $-\frac{3b}{2a}$

(C) $3b$

(D) $2a$

Q11 Let $D=a^2+b^2+c^2, a, b$ being consecutive integers and $c=ab$ then $\sqrt{D}$ is

(A) Always an even integer

(B) Sometimes an odd integer and sometimes an even integer.

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(C) Always an odd integer
(D) Always irrational

Q12 The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is
(A) 4
(B) 5
(C) 6
(D) 8

Q13 The number of real solutions of the equation, $x^2 - |x| - 12 = 0$
(A) 2
(B) 3
(C) 1
(D) 4

Q14 If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the value of $\frac{(a\alpha^2+c)}{a\alpha+b} + \frac{a\beta^2+c}{a\beta+b}$ is
(A) $\frac{b(b^2-2ac)}{4a}$
(B) $\frac{b^2-4ac}{2a}$
(C) $\frac{b(b^2-2ac)}{a^2c}$
(D) None of these

Q15 The number of real roots of the equation $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is :

[JEE Main 2020, 9 Jan (Shift 1)]
(A) 1
(B) 2
(C) 3
(D) 4

Q16 If $\log_2\left(ax^2 + x + a\right) \geq 1 \forall x \in \mathrm{R}$, then exhaustive set of values of $'a'$ is:
(A) $\left(0, 1 + \frac{\sqrt{5}}{2}\right)$
(B) $\left(1 - \frac{\sqrt{5}}{2}, 1 + \frac{\sqrt{5}}{2}\right)$
(C) $\left(0, 1 - \frac{\sqrt{5}}{2}\right)$
(D) $\left[1 + \frac{\sqrt{5}}{2}, \infty\right)$

Q17 If $b < a$, then the equation $(x - a)(x - b) = 1$ has
(A) Both roots in $[a, b]$
(B) Both roots in $(-\infty, a)$
(C) Both roots in $(b, +\infty)$
(D) One root in $(-\infty, a)$ and the other in $(b, +\infty)$

Q18 The smallest value of $x^2 - 3x + 3$ in the interval $(-3, 3/2)$ is
(A) $3/4$
(B) 5
(C) -15
(D) -20

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Answer Key

Q1	(D)
Q2	(B)
Q3	(B, C)
Q4	(C)
Q5	(B)
Q6	(C)
Q7	(D)
Q8	(C)
Q9	(A)
Q10	(A)
Q11	(C)
Q12	(C)
Q13	(A)
Q14	(C)
Q15	(A)
Q16	(D)
Q17	(D)
Q18	(A)

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